House Price Analysis – Ames Housing Dataset

Final report

1. Introduction

In this project, I used the Ames Housing dataset to analyze house prices and understand the main factors that affect them. The dataset includes information such as house size, quality, and other characteristics.

2. Cleaning Summary

The dataset had several issues that needed to be fixed before analysis. Missing values were handled by either removing columns with too many missing entries or filling them using appropriate methods such as the median or "None".

Some columns had incorrect data types and were converted to the correct format. Outliers in the SalePrice column were capped using the 99th percentile to reduce their impact. Duplicates were also checked, and no major issues were found.

All steps were combined into a reusable function called `clean_data()`.

3. Feature Engineering Summary

Several new features were created to improve the analysis. For example, **price_per_sqft** was created to compare houses more fairly, and **total_bathrooms** combined bathroom counts into one feature.

An interaction feature between quality and area was also added. In addition, SalePrice was log-transformed to reduce skewness.

Categorical features were encoded, numerical features were scaled, and highly correlated features were removed.

4. Key Findings

The analysis showed that house quality is the strongest factor affecting price. Living area also has a significant impact, as larger houses tend to be more expensive.

The original price distribution was skewed, but after applying a log transformation, it became more balanced, which improved the analysis.

5. Dataset Comparison (Bonus 🎖)

In this comparison, I analyzed two datasets: the Insurance dataset and the Ames Housing dataset with **Leen Alhossan**.

Both datasets aim to predict a numerical value, making them regression problems. In the Insurance dataset, the target variable is **charges**, while in the Ames dataset, it is **SalePrice**.

The most important feature in the Insurance dataset is **smoker**, as it strongly increases medical costs. In contrast, the Ames dataset depends on features like **Overall Qual** and **Gr Liv Area**, where prices increase gradually.

The two datasets are different in nature. The Insurance dataset focuses on health and lifestyle, while the Ames dataset focuses on physical house characteristics. As a result, the patterns are different. The Insurance dataset shows a clear gap between smokers and non-smokers, while the Ames dataset shows more stable and gradual changes.

From the results, I found that smoking has a strong and direct effect on costs, while house prices are influenced by multiple features together.

During the process, challenges such as missing values, outliers, and incorrect data types appeared. These were solved through proper data cleaning and preprocessing, which improved the accuracy of the analysis.

Overall, this comparison shows that different types of data lead to different patterns. Some datasets are influenced by one strong factor, while others depend on multiple features.

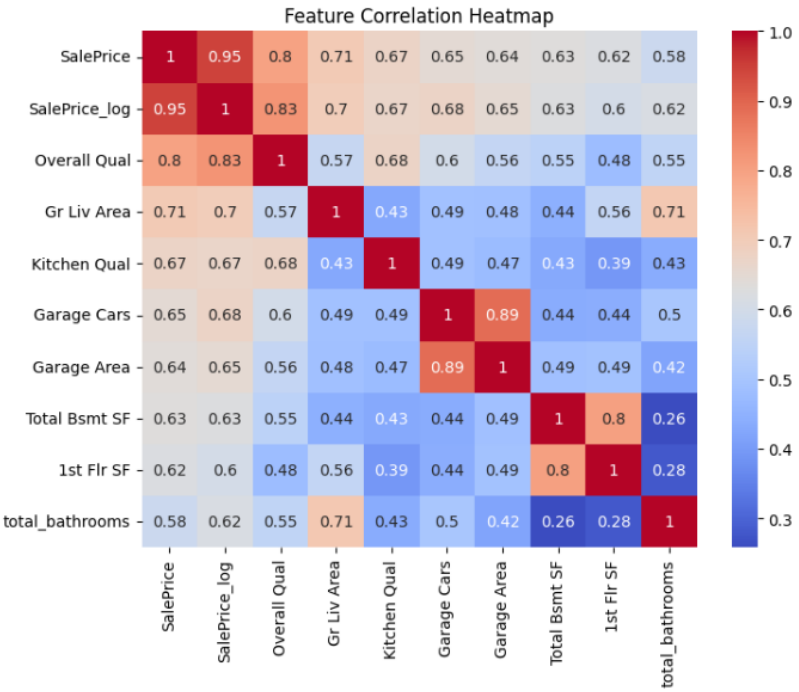
6. What I Would Do Next

If I had more time, I would:

- Build a prediction model
- Explore feature interactions further
- Analyze trends over time
- Improve and expand the dashboard for better visualization

7. best chart

The heatmap shows that Overall Quality and Gr Liv Area have the strongest correlation with SalePrice, indicating that quality and size are the main factors influencing house prices.



Conclusion

This project showed the importance of data cleaning and feature engineering in improving analysis. By preparing the data properly, I was able to identify key factors affecting house prices and gain useful insights.